Continuous Attractors in Retrosplenial Cortex

Branco Lab Meeting

Jake Laherty

June 17, 2026

Overview

- Motivation

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- (Some) Theory of continuous attractors

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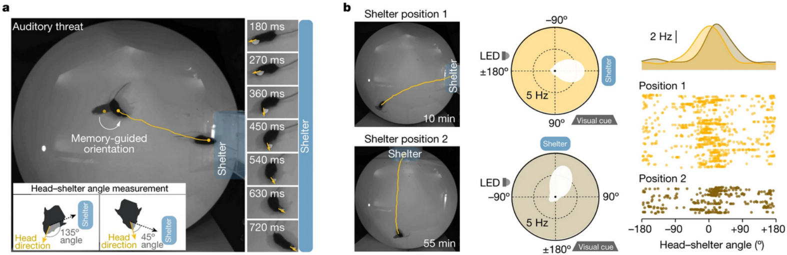

Source: Campagner et al. 2022

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The canonical model for this situation is the continuous attractor: fixed points of a dynamical system correspond to particular values of θ.

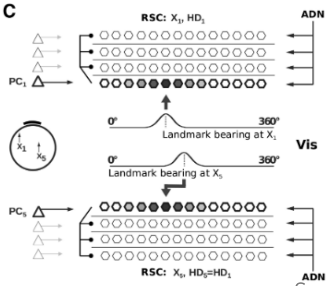
Motivation

What we were motivated by is coordinate transformations.

Suppose we have two interacting continuous attractor circuits:

- They track variables $\theta_1(t)$ and $\theta_2(t)$ whose zero is given by an external landmark (e.g. head direction, position).
- We want to compute a new variable $\varphi(t) = f(\theta_1(t), \theta_2(t))$ (e.g. head-shelter angle).

The canonical model for this is to form an intermediate circuit tuned to combinations of θ_1 and θ_2 as a basis for f; e.g. Bicanski and Burgess (2016):

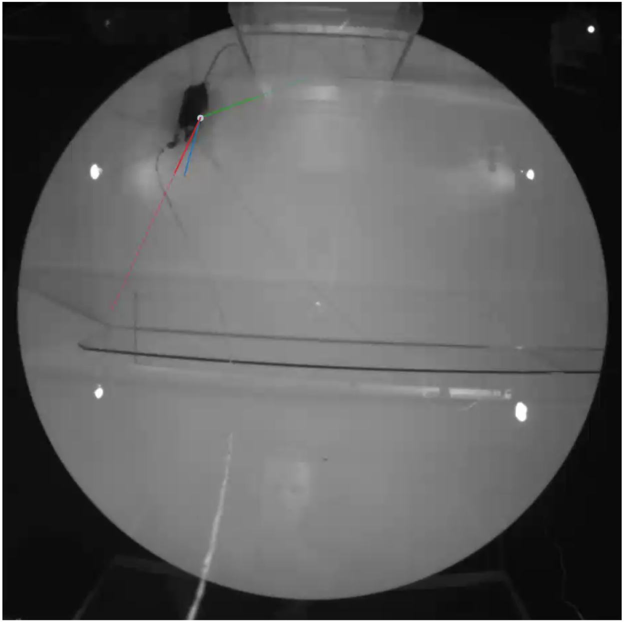


Source: Bicanski and Burgess 2016

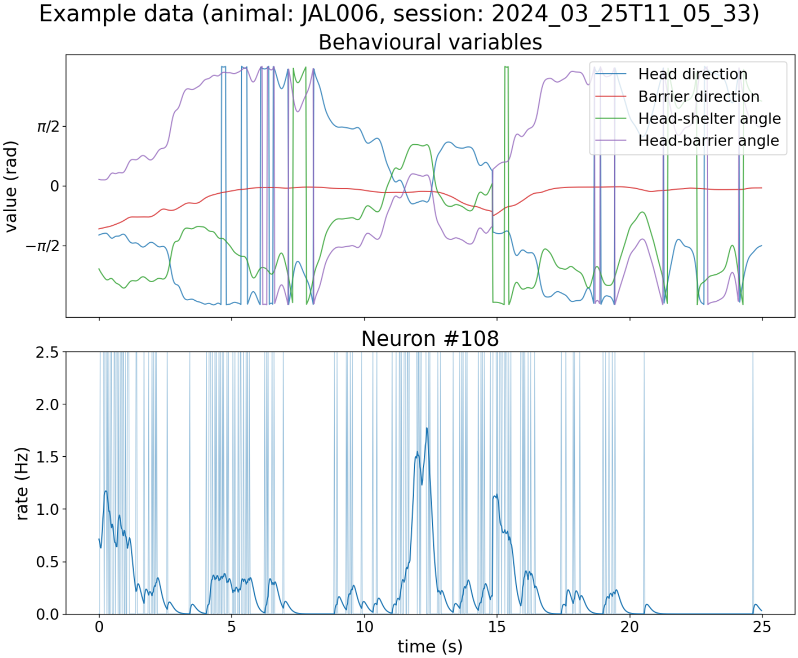
Data Analysis

Are there continuous attractors in retrosplenial cortex?

(NB: all data generously supplied by Jasmine Reggiani)



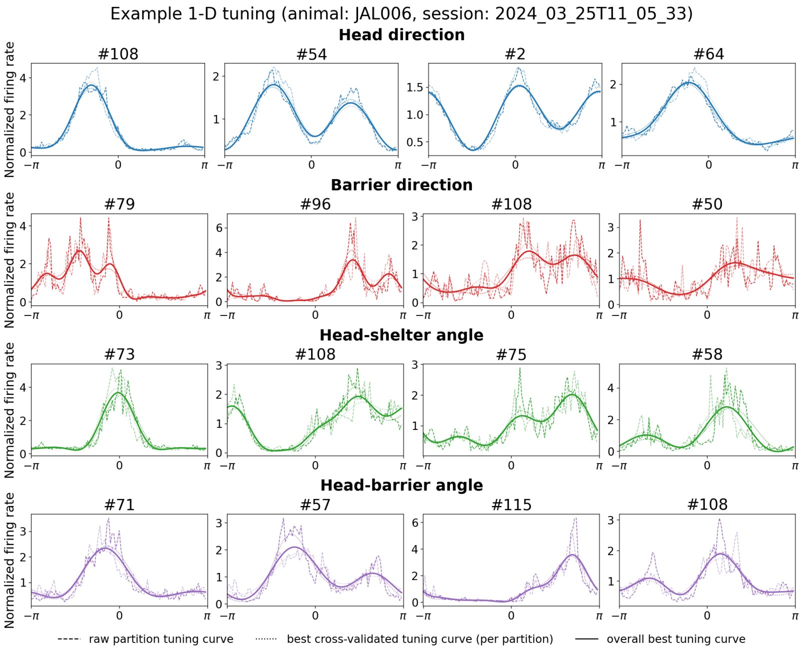
- Mice (here 3 animals, 12 sessions total) explore arena with a shelter and a barrier open at one side.
- Halfway through the session, the barrier opening is flipped to the other side.
- Neurons in retrosplenial cortex are recorded with Neuropixels ($N = 2075$ single units).
- Here we consider periods of free exploration (no escaping, no homing)
- We track: head-direction and barrier-direction (angles relative to left-center of frame); head-shelter angle and head-barrier angle (angles relative to head-direction)


Example data (animal: JAL006, session: 2024_03_25T11_05_33)
Behavioural variables
value (rad)
π/2
0
−π/2
Head direction
Barrier direction
Head-shelter angle
Head-barrier angle
Neuron #108
rate (Hz)
2.5
2.0
1.5
1.0
0.5
0.0
0
5
10
15
20
25
time (s)

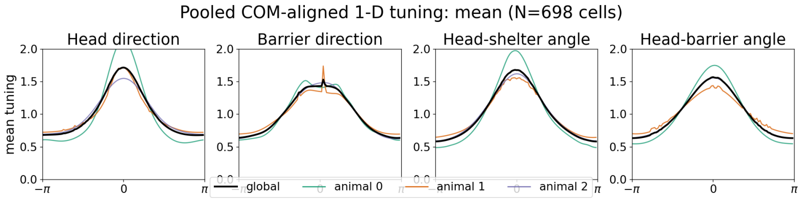
Tuning analysis

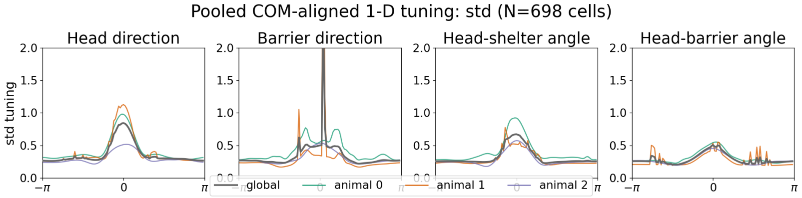
In what follows, we will be looking at tuning curves (drawing significantly from Clark et al. (2025)). These are computed by:

- Partitioning sessions into K disjoint, contiguous periods (here $K = 2$).
- Binning each behavioural variable into $D = 100$ bins and computing the average firing rate of each neuron in each bin.
- Fitting a curve to these emprical within-partition tuning curves using a periodic Gaussian Process.
- Choosing the parameters of this GP (kernel length and amplitude scales) by maximising the cross-partition Poisson likelihood.
- Fitting a curve to the whole session using the average cross-validated parameters across all partitions.



Example 1-D tuning (animal: JAL006, session: 2024_03_25T11_05_33)
Head direction
#108
#54
#2
#64
Barrier direction
#79
#96
#108
#50
Head-shelter angle
#73
#108
#75
#58
Head-barrier angle
#71
#57
#115
#108
Normalized firing rate
−π
0
π
raw partition tuning curve
best cross-validated tuning curve (per partition)
overall best tuning curve


Pooled COM-aligned 1-D tuning: mean (N=698 cells)
Head direction
Barrier direction
Head-shelter angle
Head-barrier angle
mean tuning
2.0
1.5
1.0
0.5
0.0
$-\pi$
0
π
global
animal 0
animal 1
animal 2


Pooled COM-aligned 1-D tuning: std (N=698 cells)
Head direction
Barrier direction
Head-shelter angle
Head-barrier angle
std tuning
2.0
1.5
1.0
0.5
0.0
$-\pi$
0
π
global
animal 0
animal 1
animal 2

Tuning Analysis

What does this tell us?

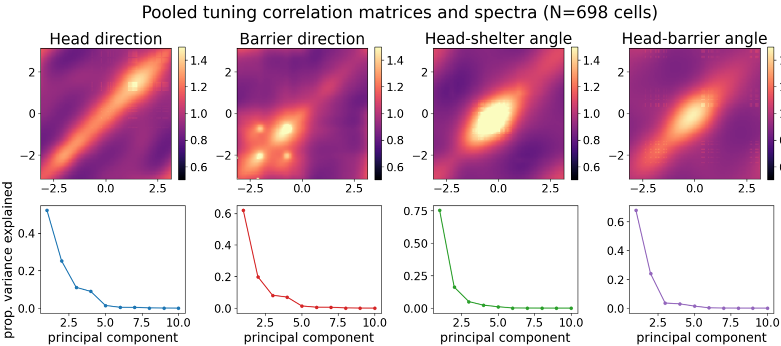
- Individual neurons can be described by tuning to behavioural variables.
- Across the population, the shape of these tuning curves is heterogeneous.

We can model this situation as a finite sample of neurons in 4 separate continuous attractor networks with disordered embeddings.

Consider the correlation matrix of tuning curves:

$$C^{\phi}(\theta, \theta') = \frac{1}{N} \sum_{i}^{N} \phi_i^*(\theta) \phi_i^*(\theta')$$

where $\phi_i^*(\theta)$ is the firing rate of neuron i as a function of the tracked variable θ (i.e. its tuning curve)



Tuning Analysis: Generative Model

We can construct a generative model of tuning curves as follows:

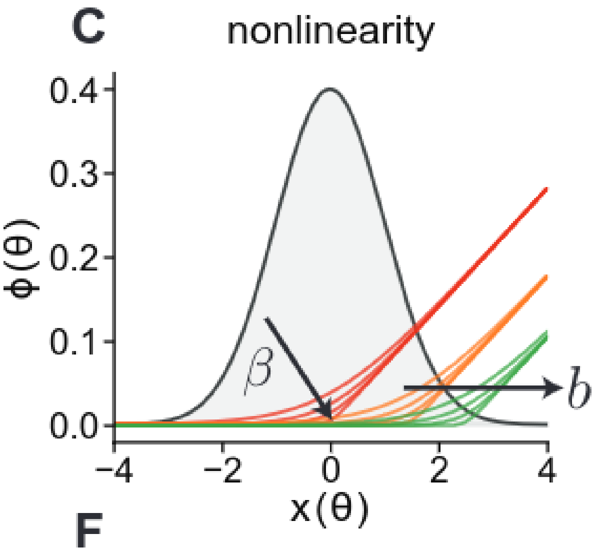
$$\phi^*(\theta) = f(x^*(\theta))$$

for some non-linear activation function f and the tuning curve of 'somatic voltage' $x^*(\theta)$. We parametrise these as:

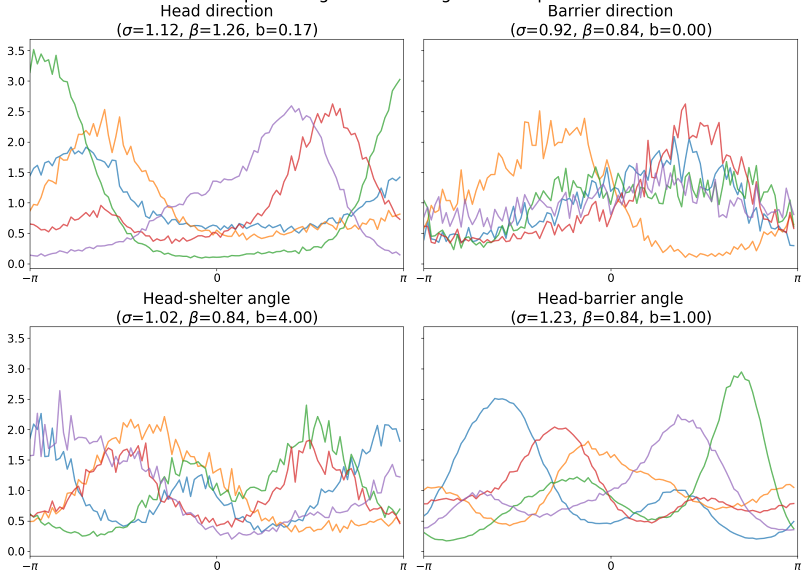
$$x^*(\theta) \sim \mathcal{GP}(0, \Gamma(\theta - \theta'; \sigma))$$

$$f(x) = \frac{1}{Z[x]} \log\left(1 + e^{\beta(x-b)}\right)$$

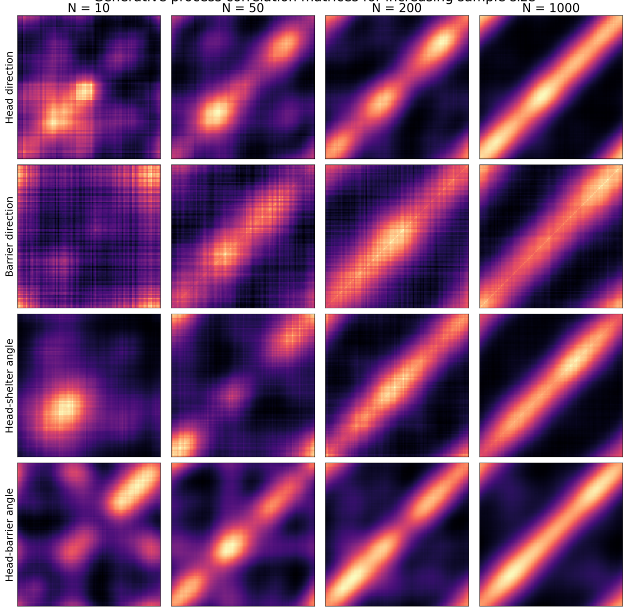
We fit the hyperparameters (σ, β, b) by minimising MSE between tuning correlation matrices sampled from the generative process and the empirical ones.

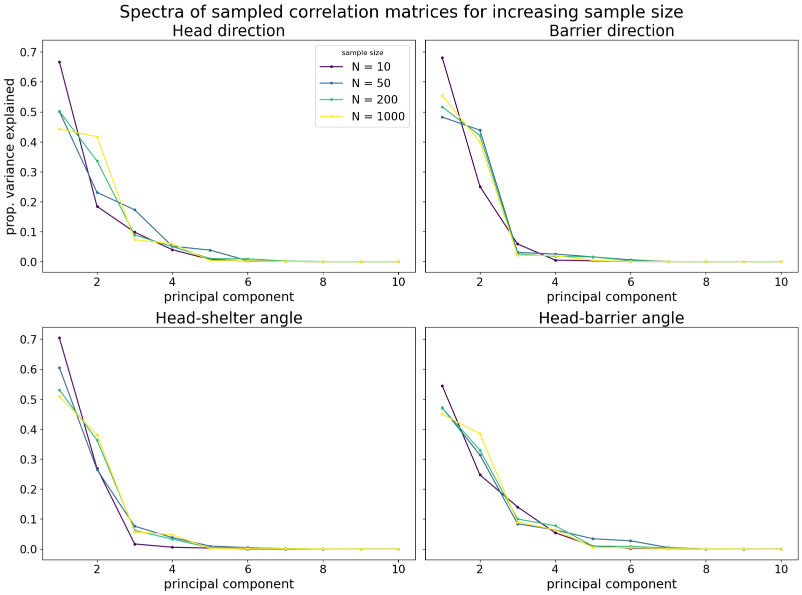

C
nonlinearity
0.4
0.3
0.2
0.1
0.0
ϕ(θ)
β
b
−4
−2
0
2
4
x(θ)
F

Sample tuning curves from generative process

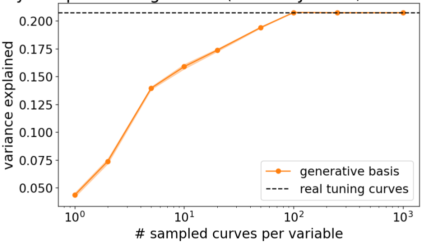


Generative process correlation matrices for increasing sample size




Spectra of sampled correlation matrices for increasing sample size
Head direction
Barrier direction
Head-shelter angle
Head-barrier angle
prop. variance explained
principal component
sample size
N = 10
N = 50
N = 200
N = 1000
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
2
4
6
8
10

Variance explained by sampled tuning curves (animal: JAL006, session: 2024_03_25T11_05_33)



Tuning Analysis: Generative Model

What does this tell us?

- Observed tunings can be modelled as a finite sample from the generative process.
- This generative process is well-behaved as network/sample size increases:
  - correlation matrices converge to something reasonable.
  - variance of overall network captured does not exceed empirical curves.

There is theory for this: we can describe our finite sample as a projection of the 'true' continuous attractor network:

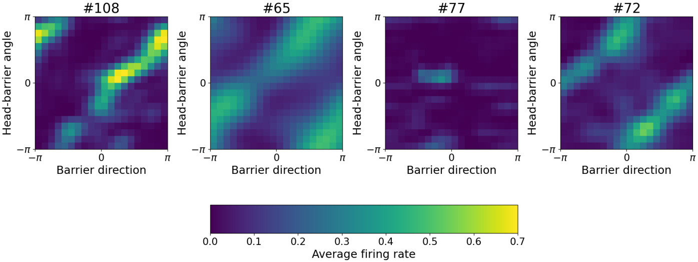
$$\frac{d}{dt}\tilde{m}(\theta, t) = -\tilde{m}(\theta, t) + G_{Q(t)}\left(\int_{-\pi}^{\pi} \mathcal{J}(\theta - \theta')\tilde{m}(\theta, t)d\theta'\right)$$

Tuning Analysis: Mixed Tuning

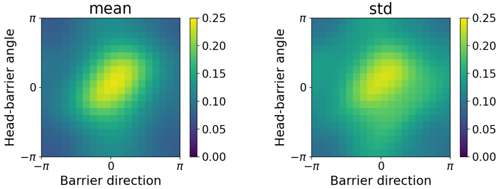
But this was all about 4 separate continuous attractors . . .
Indeed, the canonical model of coordinate transformations using continuous attractors requires neurons tuned to combinations of both variables.

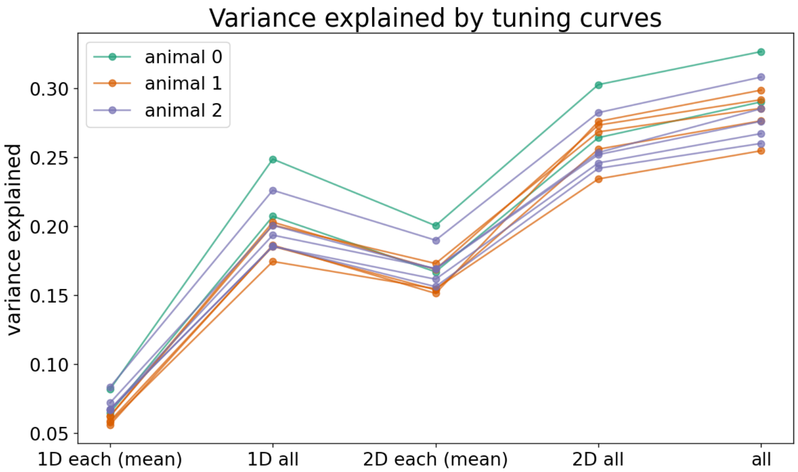
We can play the same game for joint tuning (here focusing on tuning to barrier direction and head-barrier angle)

Example 2-D tuning: Barrier Direction x Head-Barrier Angle (animal JAL006, session 2024_03_25T11_05_33)



Pooled COM-aligned 2-D tuning: Barrier Direction x Head-Barrier Angle (N=698 cells)




Variance explained by tuning curves
animal 0
animal 1
animal 2
variance explained
0.30
0.25
0.20
0.15
0.10
0.05
1D each (mean)
1D all
2D each (mean)
2D all
all

Tuning Analysis: Mixed Tuning

What does this tell us?

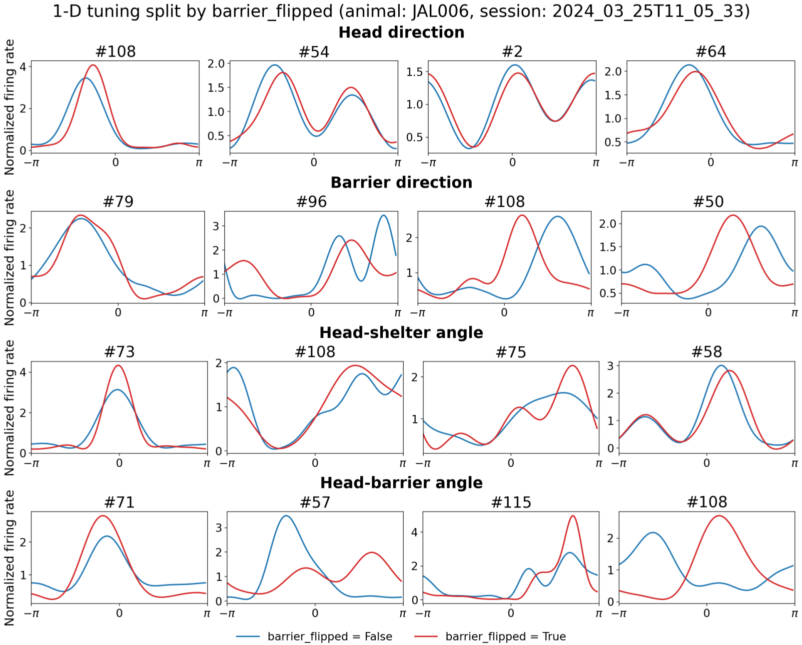
- 2D tuning curves exhibit the same heterogeneity as the 1D curves.
- 2D curves explain more variance in the whole network than the 1D curves.

This would seem to be a promising case study of a neural circuit implementing the classical coordinate transformation circuitry (albeit with disordered embeddings) . . .

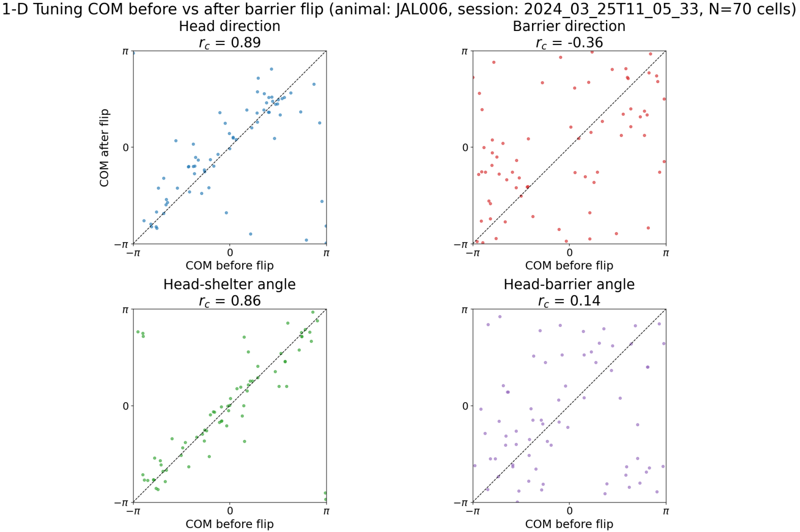
Tuning Analysis: Barrier Flip

. . . but this is not the whole story.

So far we have not considered the effect of the barrier flip of tuning profiles.
In the following, we separate sessions into the pre- and post-flip periods and compute tuning curves there separately.



1-D tuning split by barrier_flipped (animal: JAL006, session: 2024_03_25T11_05_33)
Head direction
#108
#54
#2
#64
Barrier direction
#79
#96
#108
#50
Head-shelter angle
#73
#108
#75
#58
Head-barrier angle
#71
#57
#115
#108
Normalized firing rate
$-\pi$
0
π
barrier_flipped = False
barrier_flipped = True


1-D Tuning COM before vs after barrier flip (animal: JAL006, session: 2024_03_25T11_05_33, N=70 cells)
Head direction
r_c = 0.89
Barrier direction
r_c = -0.36
Head-shelter angle
r_c = 0.86
Head-barrier angle
r_c = 0.14
COM after flip
COM before flip
π
0
$-\pi$

What does this suggest?

- The canonical model of coordinate transformations is really just one big continuous attractor: it may be a good model of variable tracking in a stable environment.
- How continuous attractors are used to compute *new* quantities in a changing environment remains an open question.
  - e.g., feed-forward models

Modelling

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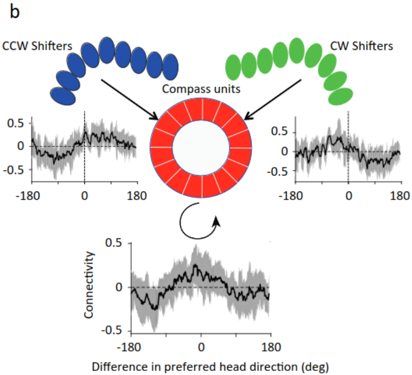
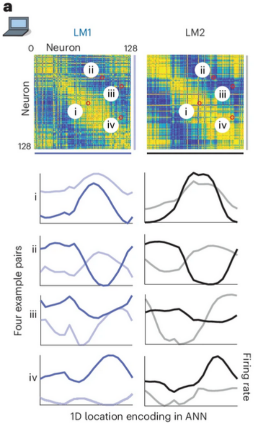
Do continuous attractors arise in task-trained RNNs?





Source: Cueva et al. 2020; Voigts et al. 2025

Modelling

From our modelling (results not shown):

- Continuous attractors do arise in certain tasks . . .

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Modelling

From our modelling (results not shown):

- Continuous attractors do arise in certain tasks ...
- but RNNs seem to favour low-rank solutions ...
- and which solutions are favoured is dependent on task construction.

(Some) Theory

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The theory for constructing continuous attractors hinges on a few components:

- Recurrent network

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- Geometry

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- Topology
- Disorder
- Geometry
- Low-rankness

Example: Line Attractor

We model a neural circuit with a linear dynamical system:

$$\frac{d}{dt} r_i(t) = \sum_{j}^{N} A_{ij} r_j(t) + h_i(t)$$

or in matrix notation:

$$\frac{d}{dt} r(t) = A r(t) + h(t)$$

For a linear system, the relevant directions are given by the *eigenvectors* of $A \to V \Lambda V^T$.

$$\begin{aligned} \frac{d}{dt} r &= V \Lambda V^T r + h \\ \frac{d}{dt} V^T r &= \Lambda V^T r + V^T h \\ \frac{d}{dt} \tilde{r} &= \Lambda \tilde{r} + \tilde{h}, \quad \tilde{r} \doteq V^T r \text{ and } \tilde{h} \doteq V^T h \end{aligned}$$

Solving in component notation gives:

$$\frac{d}{dt}\tilde{r}_i(t) = \lambda_i \tilde{r}_i(t) + \tilde{h}_i(t)$$

$$\implies \tilde{r}_i(t) = \int_0^t \tilde{h}_i(s) e^{\lambda_i (t-s)} ds$$

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Thus, we have a *manifold* of fixed points: a line attractor.

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Thus, we have a *manifold* of fixed points: a line attractor.
What do we learn?

- Eigenvalues: if $\lambda_i < 0$, the pattern $\tilde{r}_i$ dies; if $\lambda_i > 0$, the pattern $\tilde{r}_i$ runs away; if $\lambda_i = 0$, the pattern is stable.
- Symmetry: the decomposition $A = V \Lambda V^T$ is only guaranteed if A is tranpose-symmetric.
- Updates: by giving inputs along the eigenvectors, $\tilde{h}$, we can integrate them to maintain the signal.

Symmetry and Topology

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Take-away: A continuous attractor can track a variable of a particular *topology* by exploiting the right *symmetry*.

Symmetry and Topology

Classical continuous attractors for a circular variable (ring attractors) choose

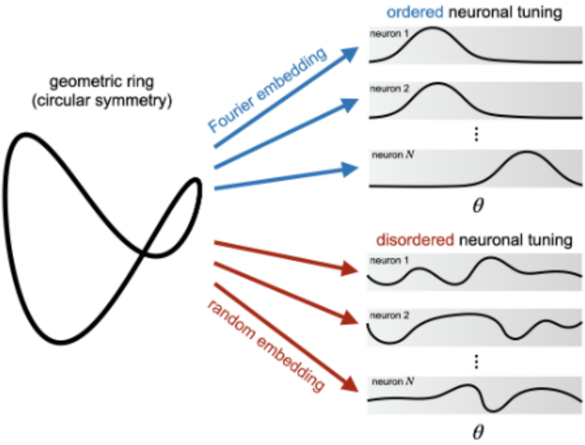
$$W(i,j) = W(i-j) \propto \cos(\theta_i - \theta_j)$$

This is an example of an *ordered embedding* of the symmetry. It predicts tuning curves that are identical up to shifts.

As we saw, this is a problematic prediction.

We can allow for more diverse tuning curves by using *disordered embeddings*. These relax the symmetry assumption: weights are no longer exactly symmetric, but retain the eigenstructure of symmetric weights.

Disorder and Geometry



Take-away: Topology is set by symmetry, or more generally by eigenstructure; *geometry* is determined by the way these symmetries are realised in the network.

Source: Clark et al. 2025

Disorder and Geometry

Large continuous attractor networks with disordered embeddings can be described by an 'effective neural network':

$$\frac{d}{dt}\tilde{m}(\theta, t) = -\tilde{m}(\theta, t) + G_{Q(t)}\left(\int_{-\pi}^{\pi} \mathcal{J}(\theta - \theta')\tilde{m}(\theta, t)d\theta'\right)$$

Disorder and Geometry

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- These averages behave like classical continuous attractor networks, with symmetric interactions: synaptic weights $W(\theta_i - \theta_j)$ are replaced by 'effective interactions' $\mathcal{J}(\theta - \theta')$, the average influence of a θ-tuned state on a θ'-tuned state.

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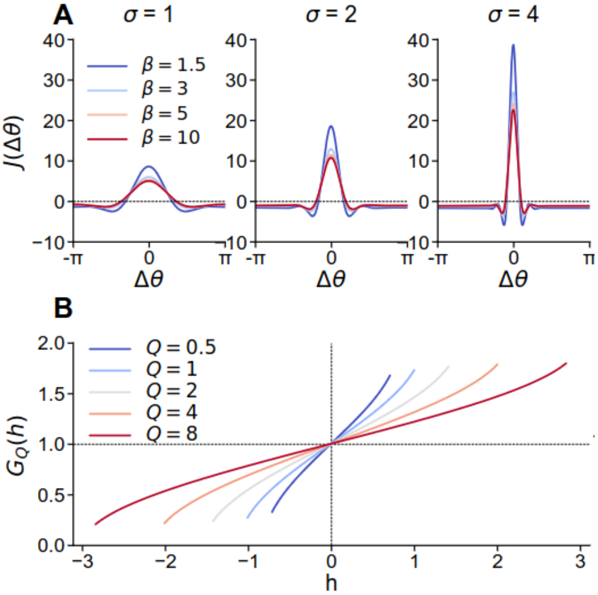
- These interactions have the interesting property that they are modulated by global activity in the network (hence the subscript $Q(t)$).
- Measuring these interactions may allow determining which continuous attractor circuits are connected.

2. Functional Connectivity



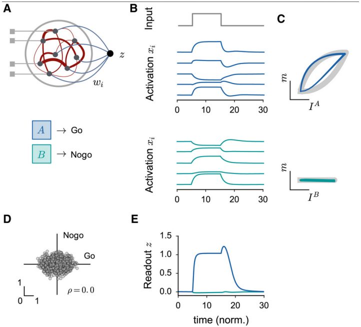
Source: Clark et al. 2025

Low-Rankness

A popular framework for studying recurrent networks uses low-rank connectivity structures (Mastrogiuseppe and Ostojic 2018):

$$W_{ij} \propto \sum_r^R m_i^{(r)} n_j^{(r)}$$

For a rank R connectivity, we can express the state of the network in terms of just R 'latent' variables.



Source: Mastrogiuseppe and Ostojic 2018

Low-Rankness

Consider the connectivity:

$$W(i,j) = \cos(\theta_i - \theta_j) = \cos(\theta_i)\cos(\theta_j) + \sin(\theta_i)\sin(\theta_j)$$

This is a low-rank network! It has rank $R = 2$ with

$$m_i^{(1)} = n_i^{(1)} = \cos(\theta_i)$$
$$m_i^{(2)} = n_i^{(2)} = \sin(\theta_i)$$

Thus the network state lies in two dimensions: $\sin\theta(t)$ and $\cos\theta(t)$

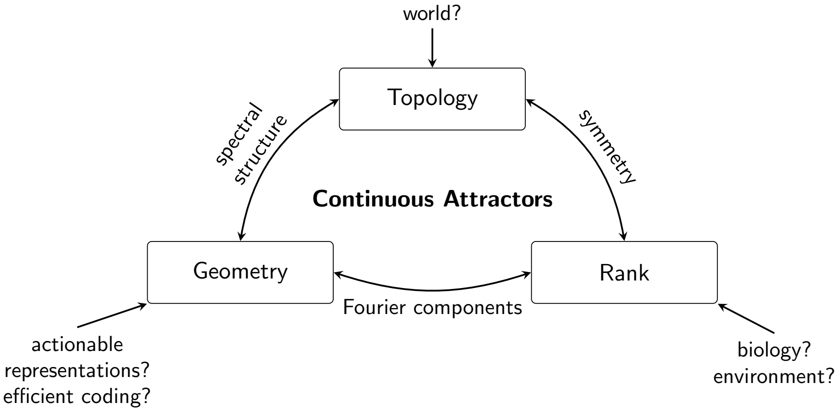
Low-Rankness

But we could equally choose (while still preserving rotational symmetry):

$$\begin{aligned} W(i,j) &= \cos(\theta_i - \theta_j) + \cos(2(\theta_i - \theta_j)) \\ &= \cos(\theta_i)\cos(\theta_j) + \sin(\theta_i)\sin(\theta_j) \\ &\quad + \cos(2\theta_i)\cos(2\theta_j) + \sin(2\theta_i)\sin(2\theta_j) \end{aligned}$$

This has rank $R = 4$ and the network state lies in four dimensions: $\sin\theta(t)$, $\cos\theta(t)$, $\sin 2\theta(t)$, and $\cos 2\theta(t)$.
Take-away: We can construct increasingly complicated tuning curves with networks of increasing rank.

Theory


world?
Topology
spectral
structure
symmetry
Continuous Attractors
Geometry
Rank
Fourier components
actionable
representations?
efficient coding?
biology?
environment?

Next steps

What to do?

Following this analysis, I have a few questions:

- When/why/how do continuous attractors emerge in task-trained RNNs?

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- How does learning interact with continuous attractors?

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An attractive framework: Statistical field theory.

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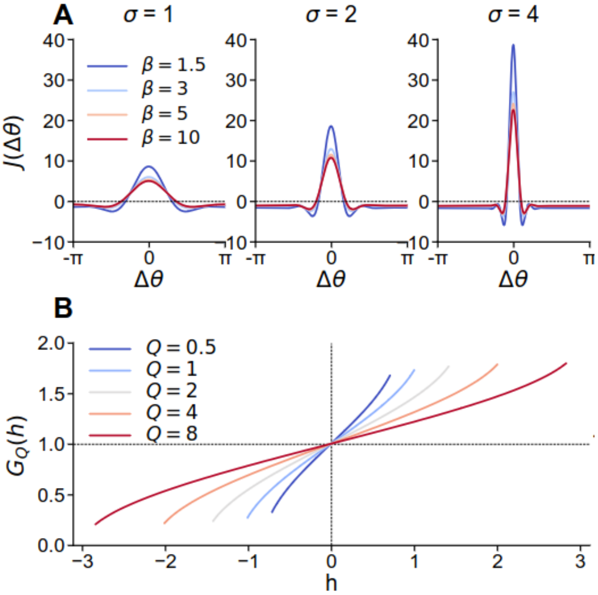

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3. Learning Dynamics

Low-rank recurrent networks are particularly amenable to analysis of learning dynamics (under gradient descent . . .)

Activity in low-rank networks is determined by the alignment between weight vectors (overlaps).
Recent work (Ger and Barak 2026) has shown that, when training under gradient descent, some of these overlaps are 'loss-visible' and some are 'loss-invisible'.

They also showed that these loss-invisible overlaps can contribute to hysteretic effects in learning trajectories.

3. Learning Dynamics

Suppose you have a rank-1 linear network that receives a scalar input $h(t)$ and produces a scalar output $y(t)$:

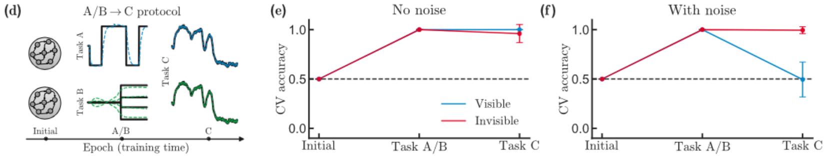
$$\frac{d}{dt}x(t) = -x(t) + \frac{1}{N}mn^T x(t) + uh(t)$$
$$y(t) = v^T x(t)$$

The connectivity vectors are (m, n, u, v)
We can write this in latent-space as:

$$\frac{d}{dt}\kappa(t) = -\kappa(t) + \begin{bmatrix} 0 & 0 \\ \frac{1}{N}n^T u & \frac{1}{N}n^T m \end{bmatrix}\kappa(t) + \begin{bmatrix} 1 \\ 0 \end{bmatrix} x(t)$$

$$y(t) = \begin{bmatrix} \frac{1}{N}v^T u & \frac{1}{N}v^T m \end{bmatrix}\kappa(t)$$

The overlaps that matter to the dynamics are $(n \cdot u, n \cdot m, v \cdot u, v \cdot m)$.
But there are other overlaps! For example $u \cdot m$.
These are the loss-visible and loss-invisible overlaps.



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  - How does the continuous attractor picture change when 'dynamic' computation is involved? How can we discern the 'static' from a 'dynamic' model of coordinate transformation?
  - Can we use low-rank networks and continuous attractors to model learning dynamics in natural settings?

References I

Ben-Yishai, R., R. L. Bar-Or, and H. Sompolinsky (1995). "Theory of orientation tuning in visual cortex". In: *Proceedings of the National Academy of Sciences* 92.9, pp. 3844–3848. DOI: 10.1073/pnas.92.9.3844.

Bicanski, A. and N. Burgess (2016). "Environmental Anchoring of Head Direction in a Computational Model of Retrosplenial Cortex". In: *Journal of Neuroscience* 36.46, pp. 11601–11618. DOI: 10.1523/JNEUROSCI.0516-16.2016.

Burak, Y. and I. R. Fiete (2009). "Accurate Path Integration in Continuous Attractor Network Models of Grid Cells". In: *PLoS Computational Biology* 5.2, e1000291. DOI: 10.1371/journal.pcbi.1000291.

Campagner, D., R. Vale, Y. L. Tan, P. Iordanidou, O. Pavón Arocas, F. Claudi, A. V. Stempel, S. Keshavarzi, R. S. Petersen, T. W. Margrie, and T. Branco (2022). "A cortico-collicular circuit for orienting to shelter during escape". In: *Nature* 613.7942, pp. 111–119. DOI: 10.1038/s41586-022-05553-9.

References II

Clark, D. G., L. F. Abbott, and H. Sompolinsky (2025). "Symmetries and Continuous Attractors in Disordered Neural Circuits". In: *bioRxiv*. DOI: 10.1101/2025.01.26.634933.

Cueva, C. J., P. Y. Wang, M. Chin, and X.-X. Wei (2020). "Emergence of functional and structural properties of the head direction system by optimization of recurrent neural networks". In: *International Conference on Learning Representations (ICLR)*. arXiv: 1912.10189.

Ger, Y. and O. Barak (2026). *Learning reveals invisible structure in low-rank RNNs*. arXiv: 2605.04115.

Langdon, C., M. Genkin, and T. A. Engel (2023). "A unifying perspective on neural manifolds and circuits for cognition". In: *Nature Reviews Neuroscience* 24, pp. 363–377. DOI: 10.1038/s41583-023-00693-x.

References III

- Mastrogiuseppe, F. and S. Ostojic (2018). "Linking Connectivity, Dynamics, and Computations in Low-Rank Recurrent Neural Networks". In: *Neuron* 99.3, pp. 609–623. DOI: 10.1016/j.neuron.2018.07.003.
- Perich, M. G., D. Narain, and J. A. Gallego (2025). "A neural manifold view of the brain". In: *Nature Neuroscience* 28.8, pp. 1582–1597. DOI: 10.1038/s41593-025-02031-z.
- Seung, H. S., D. D. Lee, B. Y. Reis, and D. W. Tank (2000). "Stability of the Memory of Eye Position in a Recurrent Network of Conductance-Based Model Neurons". In: *Neuron* 26.1, pp. 259–271. DOI: 10.1016/S0896-6273(00)81155-1.
- Voigts, J., I. Kanitscheider, N. J. Miller, E. H. S. Toloza, J. P. Newman, I. R. Fiete, and M. T. Harnett (2025). "Spatial reasoning via recurrent neural dynamics in mouse retrosplenial cortex". In: *Nature Neuroscience* 28.6, pp. 1293–1299. DOI: 10.1038/s41593-025-01944-z.